

Space initiatives have been happening since the mid 50's, many satellites and vessels have been sent into orbit only to stay in orbit. All of these objects in space are known as space junk. Space Junk is highly dangerous to any satellite and human vessels sent into space, therefore removing all of it should be and is becoming a top priority. Some of the factors to consider when removing this are the facts of how to remove it safely along with the sheer amount of debris is up there.

The amount of debris in space is an important factor because of how dangerous all the debris can be. Many countries have at least some satellites in space and as source F shows some countries have thousands of pieces of junk. All this space debris sitting up there poses the question of how are we supposed to get it all down? Every piece of debris out there poses the risk of hitting something like the ISS or some satellites. As stated in Source E [T]he small space debris was traveling at a rate of almost 17,000 mph directly towards Suomi NPP". Space debris travels at extremely high rates of speed, combine this with the fact that there are over 10,000 satellites, vessel parts and active satellites combined makes for dangerous situations, even the smallest pieces of debris can travel like a bullet through space. Since there is such a substantial number of it up there there needs to be a plan put in place to account for the amount and speed to remove it safely.

Another factor to consider is how to remove it because the debris comes from many different countries. As covered previously many different countries own the bit of the debris that litter space. And I sure doubt that these countries want other people messing with their junk, despite the fact it's junk. With so many satellites and objects belonging to places like countries to even private organizations the effort to remove it will have to be extremely organized. In source C it states "Most of what goes into space doesn't come back. Nations aren't required to remove their garbage from space...as they crash into each other, they create more tin bits of debris". Each country when not held accountable for their junk is just going to leave it up there because why spend several thousand dollars to remove that hazard when the money can go elsewhere. This leaves space organizations having to dodge the debris and private organizations have to try and step up to solve this problem. The nations should also try to be held accountable for what they are doing to the environment of space and the dangers their irresponsibility poses to the future generations of space missions. So we can all band together and pulled down more than we send up.

For years, the exploration of space has been a leading innovative concept. It introduces us to the realm outside of our planet, and how external factors as well contribute to our well-being on Earth. However, to keep efficiently doing so we must ensure that we remain clear from all contact with space debris. This debris is often caused by past human-made spacecraft, satellites, other leftover junk, etc. As a result, this can limit our space exploration posing significantly detrimental risks if collided with.

While space debris creates a dangerous path for satellites or spacecraft, it is also known to be limiting progression in the exploration of space. As stated in Document E, it can be seen that this pollution of space can not only lead to the damaging, collision, and prevention of other spacecraft in the long run, but it can also impact the Earth and our environment as a whole. When coming down to earth, if the satellite has any additional fuel, closer to Earth's altitude it can be proven that compression and friction due to the density of the region will result in the heating and breaking apart of the satellite causing more damage and harm for the Earth. Space junk would become Earth junk. Once the machinery breaks down and were to eventually return to the Pacific Ocean as stated in Document E, it could lead to the contamination of the ocean, and leading to environmental damage on the ocean/marine life present. Many forms of marine life could die as a result, and ecosystems would fail to thrive due to this external change being made to their environment. The space junk that was once preventing space exploration and posing risks to the pathways of satellites, can also contribute to one day becoming the leading cause of environmental degradation here on Earth as well.

Based on the intellect provided by Document C, it has proven to be evident that space junk can lead to becoming economically expensive, and can potentially lead to economic strains on nations and space agencies if not addressed right away. In Document C it explicitly mentions the Keisler Effect, which is when the amount of space debris made from objects crashing into one another increases, until there is no possible way to travel. The following scientific context, builds upon the idea later presented in the next paragraphs towards the beginning of the article. Document C also mentions how most of what goes into space doesn't come back, and the framework of space debris at the University of Leiden in The Netherlands has stated that nearly 9,000 satellites have been launched since the Soviet Union first sent Sputnik 1 in 1937. Crashing into each other they are creating even more debris, and space junk. In knowing that there is so much junk that is being created and multiplied, imagine the economic impact on nations once they find out that the removal of this space junk is highly expensive. If this issue is not addressed and the space junk keeps piling up, countries will have to be paying for the damage this junk is doing to other satellites, and to pay for the voluntary removal as a whole, which is also expensive. An example from Document C of a satellite being pushed back into Earth's atmosphere portrays how Canada charged \$18 million (in today's Canadian currency) for the damage caused by the Soviet Union. Calculating the expenses of trash removal, the damage caused by space junk, and then the price paid due to the collision and creation of more space junk seems like an economically draining cost to pay, almost impossible for some if the issue isn't addressed today. The excessive amount of money being spent on major trash removal will potentially lead to economic disabilities in the long run, and financial instability. While space debris creates a dangerous path for satellites or spacecraft, it is also known to be limiting progression in the

exploration of space.

Possible preventative measures that can be taken today are for this space junk to be properly disposed of right away, and for operating a clean space based on Document D it is imperative that the inevitable space junk is avoided. By ensuring that effective and innovative technology is used, we can ensure that the satellites already in space are disposable. The process of mitigating, reducing, and remediating are also effective techniques, however they need to be monitored and do not promise direct results. As encouraged by Document D, it is in the best interest of all nations and space agencies to avoid the initial cause and spread of space junk, while they can and still have the power to do so. In doing so, this will not only help limit the economic burden on economies potentially, but it will also aid in the impact that this space junk has on the environment addressing the issues presented in both Documents E and C.

Based on Documents E, C, and D it can be concluded that the most significant concerns that space agencies and nations should consider when dealing with the conflict of space debris is to look for possible prevention solutions to dispose of the space junk right away, given the fact that further pollution can be prevented as well as the limitations on space exploration, and the economic strain on nations and agencies will also be lessened. Space exploration is a crucial factor in the lives of humans, not only to aid in progressive innovations, but also helping us gain knowledge of the external factors that may impact our well-being and our planet as a whole. Having space junk that is also man-made to be interfering with such progress would be truly detrimental to society as a whole, and should be addressed considering the following pressing concerns it consists of.

do you just ever want to throw your stuff away into space and never see it again? well that's what some big business love to do, They might think it's a easy way to "clean out overstock" but in reality it's making space debris. Before we get into anything, you might ask yourself--what is space debris and why is it a problem to us ? well according to Source A O'Callaghan Jonathan states that "space junk or space debris, is any piece of machinery or debris left by humans in space" it can refer to as very big objects like a dead satellite or something very small like a paint flecks that has fallen off a rocket.

But lets get into the real problem-- In Source B, is a environment report from 2021 by ESA showing a chart satellites launched into low-earth orbit. when reading the chart you will noticed that in 2020 the commercial number sky rocketed making it the main source of space debris. thats more than what the government is throwing in space! and thats not all- let's check Source F (Chart from Mosher and Kiersz) to see which countries with the most stuff in orbit. Want to take a big guess? it's not what you think; It's Russia! or former USSR with a whopping 6,515 debris in orbit. who are you kidding, they might as well own space. And coming in 2nd and 3rd place is of course United States and China.

While others may think this isn't a big problem, In Source E (NOAA article) they say that despite the size of the debris can be hazardous to unmanned and manned spacecraft. This is because they are orbiting at extremely high velocities. This is faster than a bullet, meaning that debris can easily punch through the protective covering on satellites or spacecraft. Now if you really think about it- if a satellite or a spacecraft gets hit, where can it safely crash down to? it is in fact the earth that it's closest to and can easily land in Point Nemo (A Pacific ocean location of inaccessibility) but that is not always the case In source C (Quell article) they explain that collisions have occurred and in 1977 a crash scattered radioactive debris across Canada by the USSR. Making the Soviet Union paying 6 million Canadian Dollars (18 million today) for the damage.

So before throwing your things away, think to yourself.. will the earth benefit from this?

As human society evolves, the planet that they inhabit is usually what suffers the cost of human invention and progress. Air contamination, light pollution, sea pollution, deforestation, biodiversity loss, nuclear waste, and global warming issues make it clear that the health of the Earth is an afterthought to many of the great minds of the world. In recent centuries, though, the damage humans spread as a result of necessary creation and innovation hasn't been limited to the natural features of this planet: space, apparently, is the next stop for human littering. Space debris is taking up a small but incredibly significant portion of the atmosphere beyond the planet, and it poses many risks in the long and short term. In their effort to mitigate the space debris humans have created, space agencies and nations should be particularly concerned about the physical risks of space debris and the legal ramifications of its existence in order to effectively deal with this issue, even if the costs of eliminating space debris does not seem to be outweighed by the few immediate concerns there are.

To start, space debris poses crucial risks to both satellites and people in space. Space debris doesn't necessarily even need to be large in order to cause significant damage. An article published by NOAA, the National Oceanic and Atmospheric Administration, explains how "even the smallest of objects" are traveling "at extremely high velocities" around Earth, causing them to be incredibly dangerous on their own and even more so when orbiting in conjunction with other equally as fast pieces of debris, satellites, and rockets in their periphery: a collision could be disastrous (Source E). With private businesses launching nearly 1,300 satellites into low-Earth orbit according to a 2021 report from the European Space Agency, it's fundamentally clear that a plethora of valuable information is stored in space (Source B). The destruction of said information, often as a result of small objects which can be incredibly hazardous for "unmanned and manned" spacecrafts, makes it clear that simply letting space debris exist will only cause more problems rather than solve existing ones (Source E). For that reason, space agencies must be fundamentally aware of all the various risks space debris poses to both human lives and precious research.

To continue, space agencies and organizations should also be very familiar with the legal implications and laws that surround the topic of space exploration. Specifically, legal definitions of space responsibility and ownership must be carefully navigated and possibly changed in order to most efficiently mitigate the negative effects of space debris. As a first example, the laws surrounding space property are written in such a way that space is accessible to everyone, but the proper responsibility to keep it clean is not mandated of everyone exploring it. An article specializing in legal issues published in 2020 emphasizes how countries "aren't required to remove their garbage from space," and many don't because of the abundant costs (Source C). This implies, then, that for space organizations to take a more active role in cleaning up space, they must be mandated to do so; this, however, might take longer than expected to. The Outer Space Treaty in 1967 "declared space free for all nations to explore," but any new space laws have been very limitedly acknowledged or even ignored, being signed by "fewer than half of the world's countries" (Source C). If the nations occupying space aren't coming to a consensus about the responsibility of cleaning up the debris, then little global progress will be made in the effort to mitigate the problem of space debris. For that reason, space agencies should be aware of the legal terminology and

functions that surround their policies. Knowing that there are no laws actively get rid of space debris doesn't stop some countries, though. Even as recently as in 2014, satellite Suomi NPP was moved in light of a potential collision with debris, successfully completing mitigation maneuvers (Source E). Yet, countries that are taking action are still limited by those that do not due to space property restriction. A 2015 opinion article--undoubtedly, that may be unbiased but can be trusted as the author has the credibility of being the chief executive and cofounder of a company dedicated to eliminating space debris--explains how a "European ADR vehicle," for example, could not, "[dispose] of an Indian satellite without permission" (Source D). The United States, particularly, clearly has a surplus of resources when it comes to space exploration with more than 6,211 objects in orbit around Earth currently, so if bigger countries that can afford to spend resources on space debris removal were able to aid smaller countries and agencies like the European Space Agency, costs for them would be relatively limited as well (Source F). With this knowledge, it is incredibly clear that policies surrounding state affairs and international relationship must be addressed and improved to fully help the nations that are making an effort to eliminate space debris. Until legal changes are put into place, all of the sophisticated mitigation plans that space agencies are innovating "are real only on paper" (Source D).

In opposition to the incredible costs that space debris removal requires, many argue that the problem of space debris is not one that should concern space agencies any time soon. An article published by the Natural History Museum in London addresses this view, explaining how "none of the limited amount of space junk [in space] poses a problem" as it "doesn't pose a huge risk to [human] exploration efforts" (Source A). Looking at the issue of space debris purely in terms of what it can benefit human exploration like this, though, is slightly flawed in that human exploration is still harmed when an object flying "faster than a bullet" could potentially collide into an incredibly costly satellite that is gathering valuable research and harboring human lives (Source E). When approached in this fashion, it's clear that even though space debris clean-up requires taxpayers to pay "the cost of every single mission," it's also potentially saving a lot of money in terms of space craft and Earth damage that could cost upwards of 3 millions dollars to fix, as the Soviet Union realized when a crash of its satellites "scattered radioactive debris across Canada" (Source D; Source C). Ultimately, the costs that are associated with space debris removal should not encourage space agencies to shy away from eliminating space debris; if anything, it should encourage them to eliminate it faster.

Overall, when considering the obvious risks, legal statements, and inadvertent expenditures that surround the discussion of space debris removal, it's clear that space agencies and nations must be aware of all of these factors as they attempt to clean up space to make future missions safer.

The most important factors that space agencies and nations should consider when dealing with the problem of space debris are the cost and responsibility of dealing with these issues, the danger they pose to other space craft either manned or unmanned, and the laws surrounding space travel and the launch of space craft. according to source E space debris is traveling at a high velocity faster than a bullet, this can penetrate coverings on spacecraft. this shows that space agencies should be concerned about the dangers of space debris. According to source B in recent years a majority of satellites have been launched commercially by private businesses. space agencies should consider this factor and create regulations requiring businesses to bring satellites down once they are defunct either by themselves or paying others for the service. Source C states that nations are not required to remove their garbage, this is important because if there is no law mandating this then nothing can be done against countries who litter in space.

Space exploration was the goal for many nations in the 1900's, but those same nations did not account for the pollution their travels would leave behind. "Space junk" has become a ever-growing problem over the past century, and researchers and scientists are researching ways to mitigate the damage posed by this threat. Nations looking to fix the problem of space debris should consider the important factors of, reducing the amount of satellites being put into space, mitigation of debris through new scientific developments, and communication between nations to manage the damage of space debris.

Reducing the amount of satellites that are put into space can have a lasting impact that can mitigate the amount of space debris in the atmosphere. Over the past few decades humans have used satellites for far more than just space exploration, they have also used them for commercial uses, defense uses, civil uses, and amateur uses. This growth can be looked at positively as it shows the evolution of technology, but they point fails to see the amount of space debris being created by this excessive use of satellites. The European Space Agency, adapted these numbers into a graph that demonstrated that in 2020 alone, there were over 1,300 satellites cast into orbit. To further this point, nearly 1,200 of these satellites were of commercial uses. (Source, B) This graph demonstrates that while the evolution of technology has become more entertaining for us as individuals, we as humans fail to see the lasting impacts left from these launches. These satellites if not removed from space become dead satellites and are destined to float in space as debris until they inevitably cause damage. This damage isn't miniscule either, as the National Oceanic and Atmospheric Administration said, "Despite their size, even the smallest of objects, some of which cannot be detected by sensors, can be hazardous to unmanned and manned spacecraft." (Source, E) The authors of this source articulate that not only does it put our inventions at risk, but also the people that are manning them. Space debris is a problem that forces scientific minds, not just from one nation, but from all over the world to come together and fix this growing problem.

Almost every nation with access to space has played a part in space debris, working together these nations should be able to reduce the effects of space debris. Nations are trying to fix this problem on their own terms, failing to realize that they would be able to further reduce the risks if they were to just work together. As Mosher and Kiersz demonstrate in their graph, over 16,000 man made space items are made from just the United States, Russia, and China, not to mention the other nations that have access to space technology. (Source, F) This is an important factor as communication between nations could reduce the amount of debris put into space, by reducing the amount of objects that man throws into space to begin with. Some of these nations have already been doing research into how they themselves can change the problem of space debris through vast scientific research.

Scientists all over the world have started coming up with solutions to the problem of space debris. Active debris removal (ADR) has become an impressive scientific feat that has been furthered in recent years with the development of new technology. As Luca Rossetini writes, "Some envision an active debris removal (ADR) mission to go grab a dead satellite and remove it." (Source, D) Scientific research has been driven to help eliminate the continued problem of space debris, and if nations would work together and put their most influential minds together, they might be able to reduce and mitigate the problems of space debris.

Space debris has become a lasting problem and many factors have attributed to that, but scientists in today's day in age are trying to mitigate those problems through the most important factors of national communication, scientific development, and the reduction of satellites.